Extra Practice 1: Spatial Puzzles: Tangrams and Pentominoes

A. Skills

1. Which strategy fits each plan best? Choose from: *draw or model, guess and check, systematic list, look for a pattern.*

(a) Nadia sorts the pentominoes she finds by their longest row, so she never draws one twice. _____

(b) Ben counts squares in 1 by 1, 2 by 2 and 3 by 3 grids to predict a 6 by 6 grid. _____

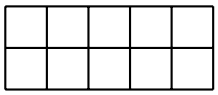
(c) Lina puts a piece in a corner. It leaves a hole, so she moves it and tries again. _____

(d) Before cutting cardboard, Theo folds a paper copy of his net to test it. _____

2. A tangram is cut from a square of card 24 cm by 24 cm. Find the area of each kind of piece. Check that the 7 pieces add up to the whole square.

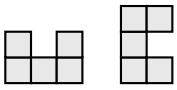
Whole square	**Small triangle**	**Medium triangle, square, parallelogram**	**Large triangle**

3. How many squares of any size are in this 2 by 5 rectangle? Make a systematic list.



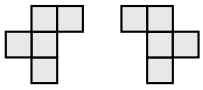
4. Is each pair the same pentomino or different? If they are the same, say whether you turn or flip it.

(a)



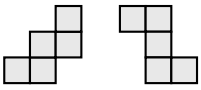
_____

(b)



_____

(c)



_____

5. Use only the **two large triangles**. Make each shape and sketch how the triangles fit.

(a) a square (b) a bigger triangle (c) a parallelogram

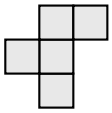
B. Applying

6. Make this square with **4** pieces: one large triangle, the medium triangle and both small triangles. (The picture is smaller than your pieces.) Draw the lines and check the area.



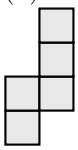
7. Does each pentomino fold into an open box? Write yes or no. If yes, shade the square that becomes the bottom.

(a) F



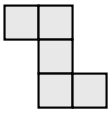
_____

(b) N



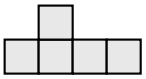
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(c) Z



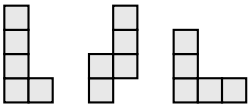
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(d) Y

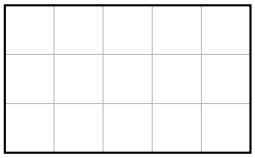


_____

8. Tile this 3 by 5 rectangle with the **L**, **N** and **V** pentominoes. Label each piece.



L **N** **V**



9. Ben says, "Each side of a large triangle is twice as long as the matching side of a small triangle, so a large triangle has twice the area of a small one." Identify and correct Ben's error.

10. Which set of tangram pieces covers exactly half of the whole tangram square?

A. one large triangle and the medium triangle

B. both large triangles and one small triangle

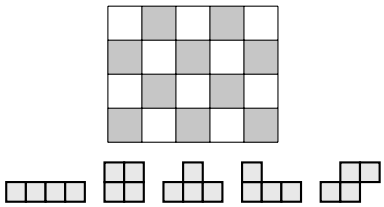
C. one large triangle, the square and the parallelogram

D. the square, the parallelogram, the medium triangle and one small triangle

11. Numerical Response The tangram on your cut-out page is a 10 cm by 10 cm square. The area of its parallelogram is _____ cm^2.

C. Stretch

12. Can the 5 different tetrominoes tile a 4 by 5 rectangle? Follow the steps to decide.



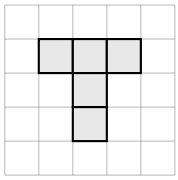
(a) Does the area rule it out? _____

(b) The rectangle is coloured like a checkerboard. How many dark squares and how many light squares are there? _____

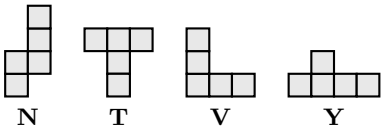
(c) Wherever you place it, the T tetromino covers 3 squares of one colour and 1 of the other. Each of the other four always covers 2 dark and 2 light. How many dark squares can the 5 pieces cover in total?

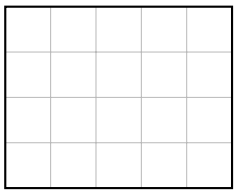
(d) So, can it be done? Explain.

13. Create a variation: a closed cube has 6 faces. The T pentomino folds into an open box. Find **every** square on the grid that you could add to the T to make a net of a closed cube. Shade them, and explain your strategy.



14. Tile this 4 by 5 rectangle with the **N**, **T**, **V** and **Y** pentominoes. Label each piece. (There is only one way, apart from turning or flipping the whole rectangle.)





Extra Practice 2: Spatial Games and Strategies

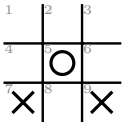
A. Skills

1. In the take-away game (take 1, 2 or 3; last counter wins), how many should you take from each pile? Write "stuck" if you cannot leave a multiple of 4.

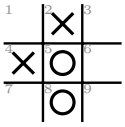
Counters on your turn	35	22	33	40
Take				

2. Use the tic-tac-toe checklist (win, block, fork, centre). Where should the player whose turn it is play? Give the square number and the reason.

(a) O to play.



(b) X to play.



3. Use the tower-puzzle rule, fewest moves $= 2^n - 1$.

(a) How many moves does a 12-disc tower take?

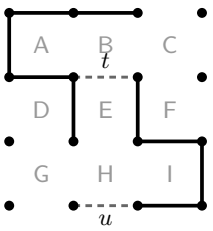
(b) A tower takes 2047 moves. How many discs does it have? Use guess and check.

4. In the hidden-ships game, you hit the other player's 2-ship at B2. Your calls at B3 and A2 were misses.

(a) Make a systematic list of every place the 2-ship could be. Cross out the ones that are eliminated.

(b) What is the greatest number of calls you still need to sink it? Explain.

5. Look at this dots-and-boxes game. The boxes are named A to I.



(a) Which boxes have exactly 3 sides? Which lines take them?

(b) Which boxes have exactly 2 sides?

(c) After you take A and I, is the dashed line t safe? Is u safe?

B. Applying

6. A take-away game: take 1, 2, 3 or 4 counters, and the player who takes the **last** counter **loses**.

(a) Work backward. Complete the table.

Counters on your turn	1	2	3	4	5	6	7	8	9	10	11	12
Win or lose?												

(b) Which numbers should you leave for the other player?

(c) The pile has 20 counters and you go first. How many should you take?

7. In a variation of the take-away game, each player may take 1 to 6 counters, and the last counter wins. The pile has 50 counters.

(a) Which numbers should you leave? Explain.

(b) Should you go first? If so, how many should you take?

8. Here is a tic-tac-toe game between Liam (X) and Hana (O): X plays 5, O plays 1, X plays 9, O plays 2, X plays 3. Liam wins on his next turn.

(a) Why did Liam have to play 3?

(b) Why can't Hana stop Liam now?

(c) Which of Hana's moves was the mistake? Where should she have played?

9. Variation of dots and boxes: each box is worth 1 point, but the 4 corner boxes (A, C, G and I on a 3×3 board) are worth 2 points each. The player with more points wins. How should the strategy change?

10. A tower puzzle at a games café has 6 discs. A player makes 1 move every 2 seconds and never makes a mistake. How long does it take? Give minutes and seconds.

11. In the hidden-ships game, you hit a 3-ship at D3. Your calls at D2, D4 and C3 were all misses. Which square should you call next?

A. E3 **B.** D5 **C.** C2 **D.** B3

12. Numerical Response A tower puzzle has 9 discs. A player has made 200 moves without a mistake. The number of moves still needed to finish is _____.

C. Stretch

13. The two-pile game. There are two piles of counters. On your turn, take as many counters as you like (at least 1) from **one** pile. The player who takes the last counter wins.

(a) The piles have 5 and 3 counters, and you go first. Show a move that makes the piles equal.

(b) Explain why you can always win after making the piles equal.

(c) The piles start at 4 and 4. Should you go first or second?

14. In the tower-puzzle variation where a disc may move only to a **neighbouring** peg (never straight between A and C), 1 disc takes 2 moves and 2 discs take 8 moves.

(a) To move 3 discs from A to C: move the top 2 from A to C, the big disc from A to B, the top 2 from C back to A, the big disc from B to C, then the top 2 from A to C. How many moves is that?

(b) Look at 2, 8, 26. Find a pattern and predict the moves for 4 discs.

15. Variation of the hidden-ships game: ships may also lie diagonally.

(a) Does the checkerboard hunt still find every 2-ship for sure? Explain with a sketch.

(b) After a hit, how many squares next to it should you now try?

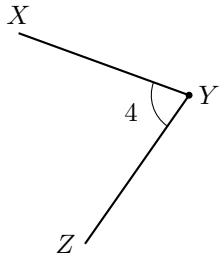
Extra Practice 3: Measuring, Drawing and Classifying Angles

A. Skills

1. Write the type of each angle: acute, right, obtuse, straight or reflex.

Angle	7°	152°	270°	90°	186°	44°
Type						

2. Use the picture.



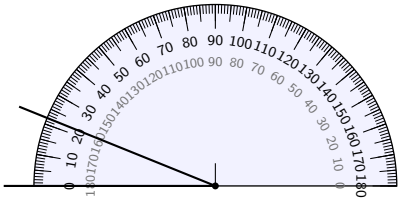
(a) Name $\angle 4$ using three letters. _____

(b) Name its vertex and its arms.

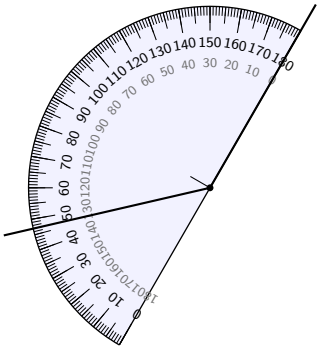
_____

(c) Can this angle also be called $\angle Y$? Explain.

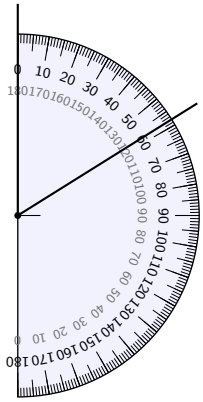
3. Read each protractor. Give the size of the angle.



(a) _____



(b) _____



(c) _____

4. Find the reflex angle that goes with each smaller angle.

(a) 12° (b) 100° (c) 155°

5. Find the sum of the interior angles of a polygon with

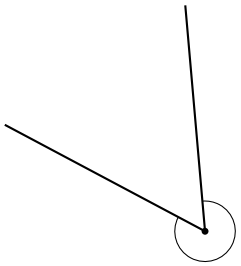
(a) 11 sides (b) 16 sides

B. Applying

6. Use your protractor. (The picture is full size.)

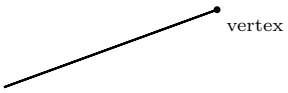
(a) Measure the smaller angle between the arms. _____

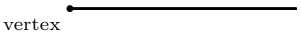
(b) Find the marked reflex angle.



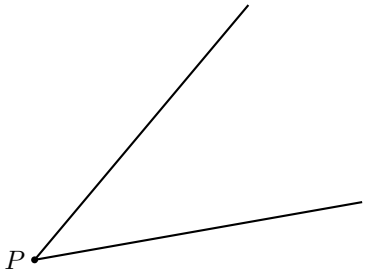
7. Use a protractor and a ruler to draw each angle on the arm given. Mark and label it.

(a) 75° (b) 155°





8. Jaxon looks at these two angles. He says $\angle P$ is bigger because its arms are longer. (The pictures are full size.)

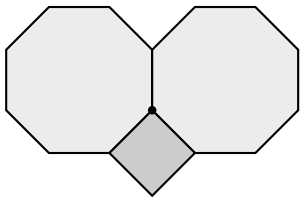




Is Jaxon right? Measure both angles to explain.

9. A kitchen floor uses tiles shaped like regular octagons, with small square tiles in the gaps. At each point, two octagons and one square meet.

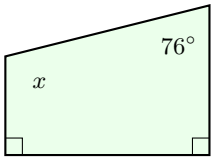
(a) Find the size of each angle of a regular octagon.



(b) Show that the tiles fit around a point with no gap.

10. Find the missing angle x.

(a) A building lot in Strathmore has two square corners and one corner of 76°.



(b) A hexagon has five angles of 130°, 118°, 125°, 112° and 120°. Find the sixth angle.

11. Lena is measuring an angle that is clearly obtuse. Where the second arm crosses her protractor, the two scales show 38 and 142. The angle is

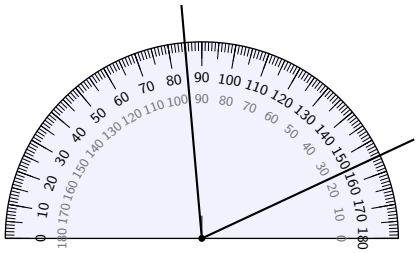
A. 38° **B.** 52° **C.** 142° **D.** 218°

12. Numerical Response A regular polygon has 24 sides. Each of its interior angles is _____°.

C. Stretch

13. The interior angles of a polygon add up to 2880°. How many sides does it have? Check your answer.

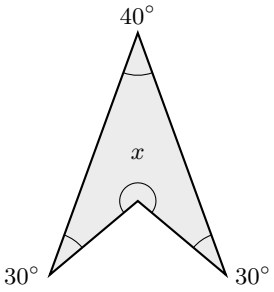
14. Neither arm of this angle is on a 0. Find the size of the angle anyway. Explain your method.



15. This arrowhead shape is a quadrilateral with a reflex angle. Its other angles are 40° at the tip and 30° at each wing.

(a) Does the rule $(n-2)\times 180°$ still work? Use it to find x.

(b) What type of angle is x? _____



Extra Practice 4: Estimating Angles with Referents, and True Bearings

A. Skills

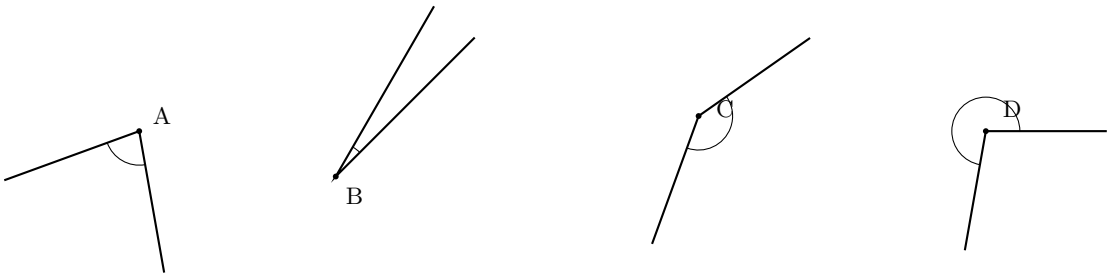
1. The minute hand of a clock turns as time passes. How many degrees does it turn in each time?

(a) 5 min _____ (b) 15 min _____ (c) 10 min _____ (d) 20 min _____ (e) 30 min _____ (f) 45 min _____

2. Build each angle from referents. Then name its type.

(a) 52.5° (b) 165° (c) 240° (d) 315°

3. Estimate each angle. Then measure it with a protractor.



Angle	A	B	C	D
Estimate				
Measure				

4. Complete the table.

Direction	S	NE	WNW	ESE		
True bearing					270°	067.5°

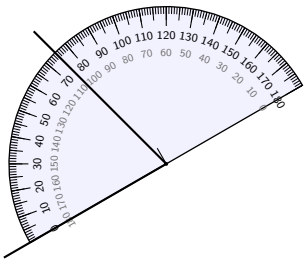
5. Find the bearing back for each bearing.

Bearing	020°	095°	180°	263°	345°
Bearing back					

B. Applying

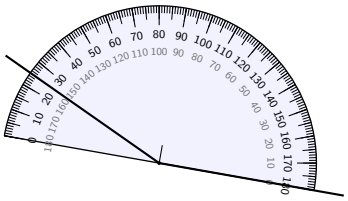
6. Estimate each angle. Then read the protractor and write the angle.

(a)



Estimate: _____ Angle: _____

(b)



Estimate: _____ Angle: _____

7. Write each direction as a true bearing.

(a) A drone flies 40° south of west. (b) A hunter walks 15° east of south. (c) A ferry sails 70° east of north.

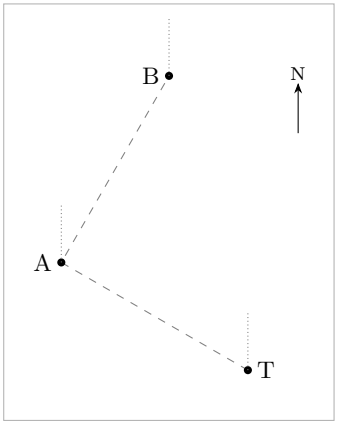
8. A bush pilot based in a northern Alberta town (T) flies supplies to two lakes, A and B. Use a protractor.

(a) Bearing from T to A _____

(b) Bearing from A to B _____

(c) Bearing from B straight home to T _____

(d) Another day the pilot flies only to A. On what bearing does she fly home from A? Use the rule.



9. Noor leaves a dock and paddles 2 km on a bearing of 070°. Then she paddles 3 km on a bearing of 160°.

(a) Make a scale drawing: 1 cm for 1 km. Start with a north line at the dock.

(b) Draw a line from where she ends back to the dock. Measure the bearing she must paddle to go straight back, and the distance.

10. A hiker walks on a bearing of 290°. Which describes her direction?

A. a little north of west

B. a little south of west

C. a little north of east

D. a little east of north

11. Numerical Response A sailboat on Lake Minnewanka is heading on a bearing of 250°. The skipper turns clockwise until the boat heads WNW. The angle the boat turns, in degrees, is _____.

C. Stretch

12. Liam is told to walk "30° west of north". He writes the bearing 030°.

(a) Make a sketch. What is his mistake?

(b) What is the correct bearing? How many degrees apart are his direction and the correct one?

13. The 16-point compass has N, NNE, NE, ENE, E, and so on, all around.

(a) Explain why the points are 22.5° apart.

(b) What bearing is exactly halfway between N and NNE?

(c) How many degrees is it clockwise from ENE to SSE? What type of angle is that?

14. At 3:00 the angle between the hands of a clock is 90°. Ava says the angle at 3:30 is also 90°, because the hour hand still points at the 3.

(a) Where is the hour hand at 3:30, really?

(b) Find the angle between the hands at 3:30.

(c) Use the same idea to find the angle at 9:30.

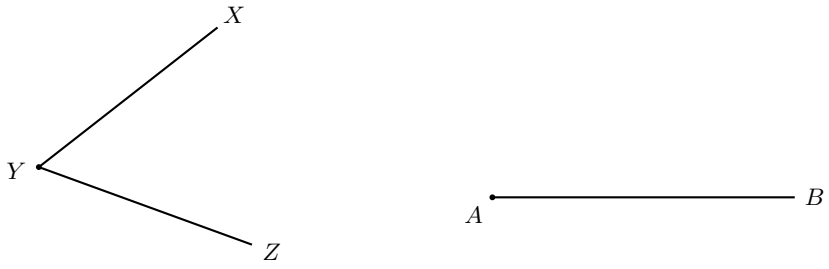
Extra Practice 5: Copying and Bisecting Angles; Perpendicular Lines

A. Skills

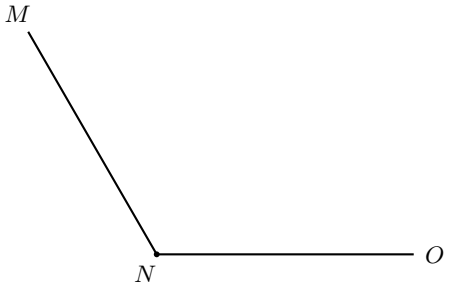
1. Each angle is bisected. Complete the table.

	(a)	(b)	(c)	(d)	(e)	(f)
Whole angle	52°	118°	25°	170°		
Each half					41°	26.5°

2. Measure $\angle XYZ$ with a protractor. Then use the protractor to copy it onto ray AB, with its vertex at A.



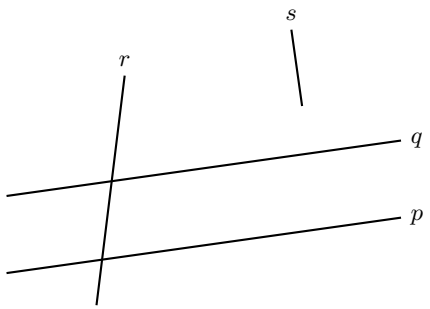
3. Bisect $\angle MNO$ with a compass and straightedge. Measure each half. What should each half be?



4. Construct the perpendicular bisector of segment EF. Label the midpoint M. How long should EM be?



5. Sort each pair of lines as parallel, perpendicular or neither. Give a reason.



Pair	Sort	Reason
p, q		
p, r		
q, s		
p, s		

B. Applying

6. A reflex angle of 250° is bisected.
(a) What is the size of each half?

(b) What type of angle is each half?

7. A framer checks a wall stud. It makes an angle of 91.5° with the floor plate on one side.
(a) Is the stud perpendicular to the floor plate? Explain.

(b) What angle does it make on the other side?

(c) By how many degrees must the stud be moved?

8. Noor is making a frame shaped like an equilateral triangle (3 equal sides and 3 equal corners).
(a) Find the corner angle.

(b) Find the cut angle at each end of a piece, and the mitre saw setting.

9. In a bay window, two walls meet at a corner of 150°. Two pieces of baseboard meet in this corner with a mitre joint that bisects the corner. What angle does each joint line make with its wall?

10. A gate is a rectangle 1.8 m wide and 1.2 m high. A brace runs from one corner to the opposite corner. Dev says the brace makes 45° with the top rail, "because it bisects the corner." Is he right? Explain.

11. Two lines meet at an angle of 88°. Which statement is true?

A. They are parallel, because they are close to 90°.

B. They are perpendicular, because 88° rounds to 90°.

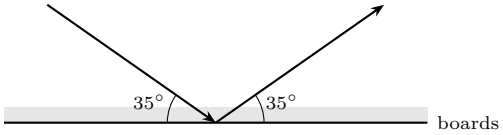
C. They are neither, because they meet but not at 90°.

D. They are neither, because they never meet.

12. Numerical Response An angle of 157° is bisected. Each new angle, to the nearest tenth of a degree, is _____°.

C. Stretch

13. A hockey player passes the puck off the boards to a teammate. The puck hits the boards at an angle of 35° and bounces off at 35° on the other side.

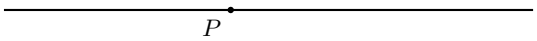


(a) What is the angle between the path in and the path out?

(b) Draw a line perpendicular to the boards where the puck hits. What angle does it make with each path?

(c) What does the perpendicular line do to the angle in (a)?

14. Use only a compass and a straightedge to construct a 45° angle with its vertex at P. (Hint: first make a 90° angle.) Describe your steps.



15. A flagpole is checked with a level. From the front it makes 90° with the ground. From the side it makes 87° with the ground.

(a) Is the pole perpendicular to the ground? Explain.

(b) How many degrees must it be moved, and in which direction of checking?

Extra Practice 6: Angle Partners: Complementary, Supplementary and Opposite Angles

A. Skills

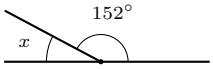
1. Find the complement and the supplement of each angle. Write "none" if it does not exist.

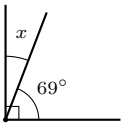
Angle	Complement	Supplement
(a) 33°		
(b) 81°		
(c) 12.5°		
(d) 140°		

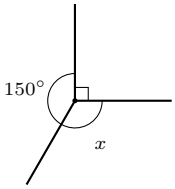
2. Each pair of angles is adjacent. Write *complementary*, *supplementary* or *neither*.

(a) 61° and 29° _____ (b) 99° and 81° _____ (c) 40° and 40° _____ (d) 123° and 47° _____

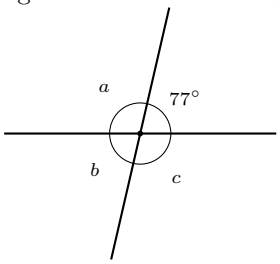
3. Find each missing angle x.

(a) 

(b) 

(c) 

4. Two straight lines cross. Find a, b and c. Give a reason for each.



B. Applying

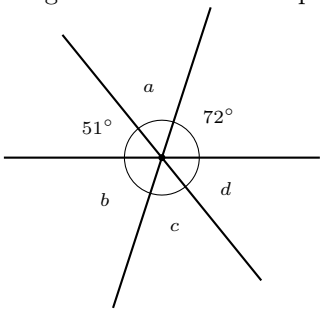
5. The complement of an angle is 14°. Find the angle and its supplement.

6. A door is hinged on a straight wall. When it is open, it makes a 65° angle with the wall on one side. What angle does it make with the wall on the other side? Which angle fact did you use?

7. The hands of a clock meet at the centre. There are 30° between each pair of hour marks.
(a) At 5:00, what is the smaller angle between the hands? What is the reflex angle?

(b) At 2:00, what is the smaller angle? What is the reflex angle?

8. Three straight lines cross at one point. Find a, b, c and d. Give a reason for each.



9. Liam says, "A 30° angle and a 60° angle can only be complementary if they are next to each other." Is he right? Explain.

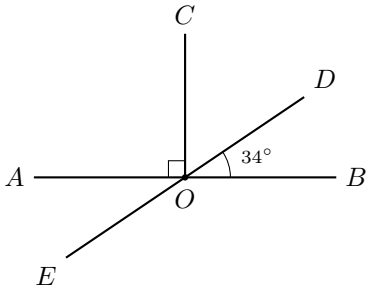
10. In a mini-golf hole, the ball rolls into a straight wall at a 27° angle and bounces off at a 27° angle on the other side. What is the angle p between the ball's path in and its path out?

11. Which angle has a supplement but **no** complement?
A. 30° **B.** 75° **C.** 130° **D.** 200°

12. Numerical Response Two straight roads cross. One of the angles between them is 57°. The size of an obtuse angle at this crossing, in degrees, is _____.

C. Stretch

13. AB is a straight line through O. $\angle AOC$ is a right angle. DE is a straight line through O, and $\angle BOD = 34°$. Find each angle and give a reason.



(a) $\angle COD$ (b) $\angle AOE$ (c) $\angle COE$ (d) $\angle BOE$

14. Two angles are supplementary. One angle is 40° bigger than the other. Find both angles. (Hint: try a pair that adds to 180°, then adjust.)

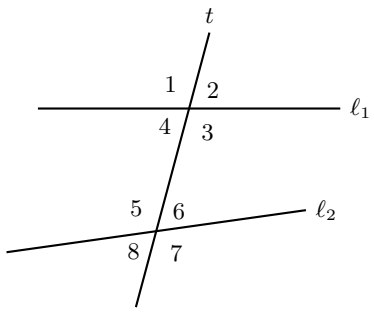
15. Two straight lines cross, and one of the four angles is a right angle. Show that the other three angles are also right angles. What do we call two lines like this?

Extra Practice 7: Transversals: Naming the Angle Pairs

In this practice the angles are numbered the usual way: 1, 2, 3, 4 clockwise from the upper left at the top crossing, and 5, 6, 7, 8 clockwise from the upper left at the bottom crossing.

A. Skills

1. Use this diagram. The lines ℓ_1 and ℓ_2 are not parallel.



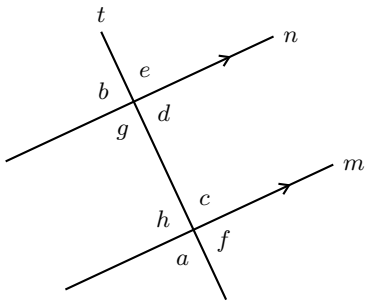
(a) List all four pairs of corresponding angles.

(b) List both pairs of alternate interior angles.

(c) List both pairs of alternate exterior angles.

(d) List both pairs of exterior angles on the same side of t.

2. Lines m and n are parallel, and t is a transversal. The angles are labelled with letters. Name each pair.



(a) $\angle c$ and $\angle g$

(b) $\angle h$ and $\angle g$

(c) $\angle a$ and $\angle e$

(d) $\angle f$ and $\angle d$

(e) $\angle c$ and $\angle b$

3. True or false? If false, fix the statement so it is true.

(a) A transversal crosses two lines at the same point.

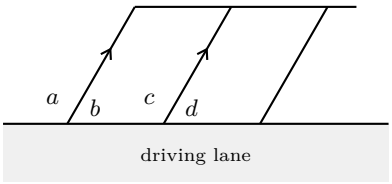
(b) Interior angles are between the two lines.

(c) Corresponding angles are at the same crossing.

(d) When two lines are not parallel, their alternate interior angles are usually not equal.

B. Applying

4. The painted lines of a row of angled parking stalls are parallel. They meet the edge of the driving lane.



(a) For $\angle a$, $\angle b$, $\angle c$ and $\angle d$, what are the two lines and what is the transversal?

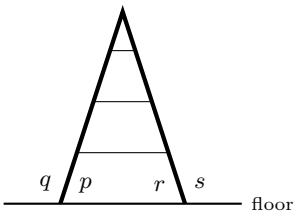
(b) Name each pair.

$\angle b$ and $\angle d$: _____

$\angle b$ and $\angle c$: _____

$\angle a$ and $\angle d$: _____

5. The two legs of a stepladder stand on a level floor. The legs meet at the top.



Use the two legs as "the two lines" and the floor as the transversal.

(a) Name the pair $\angle p$ and $\angle r$. _____

(b) Name the pair $\angle q$ and $\angle s$. _____

(c) Each leg makes a $72°$ angle with the floor on the inside, so $\angle p = \angle r = 72°$. Find $\angle p + \angle r$. Darnell says interior angles on the same side always add to $180°$. Explain why that is not true here.

6. Which sentence describes **alternate exterior** angles?

A. Both are between the two lines, on opposite sides of the transversal.

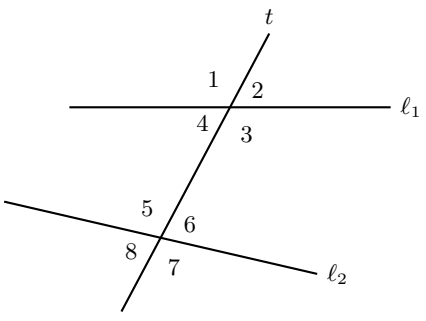
B. Both are outside the two lines, on opposite sides of the transversal.

C. Both are outside the two lines, on the same side of the transversal.

D. They are in the same position at the two crossings.

7. Numerical Response Two lines are cut by a transversal and the angles are numbered the usual way. The lines are **not** parallel. $\angle 3 = 123°$ and $\angle 8 = 71°$. The value of $\angle 1 + \angle 6$, in degrees, is _____.

8. In this diagram, ℓ_1 and ℓ_2 are **not** parallel. $\angle 4 = 62°$ and $\angle 5 = 105°$.

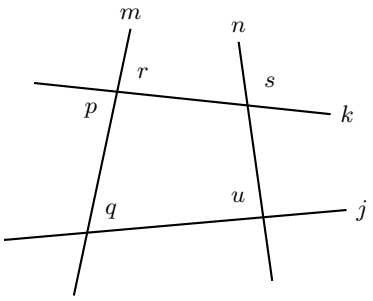


(a) Find the other six angles. Give a reason for each.

(b) Name the pair $\angle 1$ and $\angle 5$. Are they equal?

C. Stretch

9. Four lines j, k, m and n make a # shape. None of them are parallel.



For each pair, name the transversal, the two lines, and the pair.

Pair	Transversal	Two lines	Name
p & q			
r & s			
q & u			

10. Kim says, "Alternate interior angles are always equal, because they make a Z." Use your answers to Question 8 to show that Kim is wrong. Then rewrite her sentence so that it is true.

11. Pat named four pairs in a diagram numbered the usual way. Two of Pat's answers are wrong. Find them and correct them.

(i) $\angle 2$ and $\angle 6$: corresponding

(ii) $\angle 3$ and $\angle 6$: alternate interior

(iii) $\angle 1$ and $\angle 7$: alternate exterior

(iv) $\angle 4$ and $\angle 8$: exterior on the same side

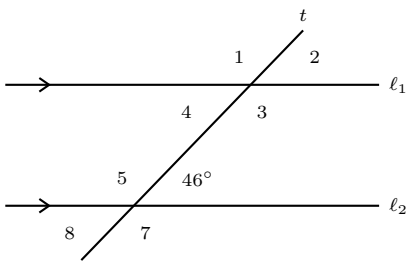
12. With 8 angles there are 28 different pairs. Make a systematic list: how many pairs are vertically opposite, how many make a straight line at one crossing, and how many have each name from this lesson? How many pairs have no special name, and which are they?

Extra Practice 8: Parallel Lines and Transversals: Finding Missing Angles

In every diagram, angles 1 to 4 are at the top crossing and 5 to 8 at the bottom crossing, numbered clockwise from the upper-left. Arrow marks (>) show parallel lines.

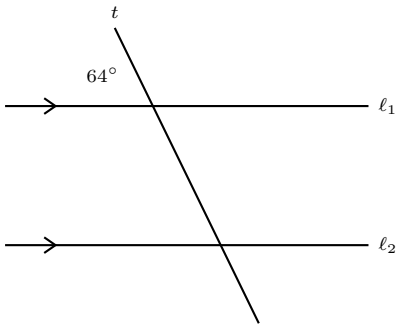
A. Skills

1. $\ell_1 \parallel \ell_2$ and $\angle 6 = 46°$. Find each angle and give a reason.

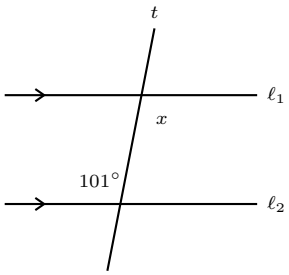


Angle	Size	Reason
(a) $\angle 2$		
(b) $\angle 4$		
(c) $\angle 8$		
(d) $\angle 5$		
(e) $\angle 3$		

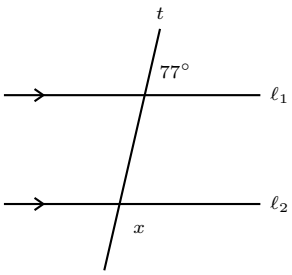
2. $\ell_1 \parallel \ell_2$ and $\angle 1 = 64°$. Write the size of every other angle in its corner of the diagram.



3. $\ell_1 \parallel \ell_2$. Name the pair of angles, write an equation, and solve for x.



(a)



(b)

4. True or false? If false, fix the sentence.

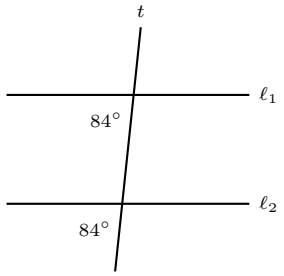
(a) When two lines are parallel, alternate exterior angles are equal. _____

(b) When two lines are parallel, interior angles on the same side of the transversal are equal. _____

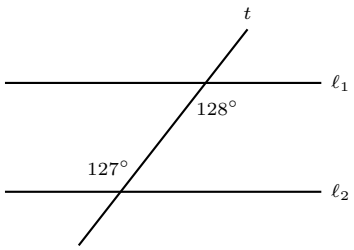
(c) Vertically opposite angles are equal only when the lines are parallel. _____

(d) If a transversal is perpendicular to one of two parallel lines, it is perpendicular to the other one too. _____

5. Are ℓ_1 and ℓ_2 parallel? Give the rule you used. (Not drawn to scale.)



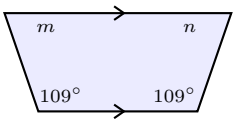
(a)



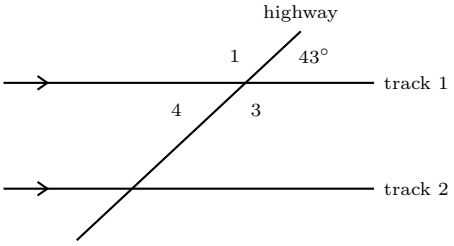
(b)

B. Applying

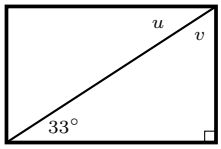
6. The end of a cattle water trough is a trapezoid. The top edge is parallel to the bottom. Each angle at the bottom, inside the trough, is 109°. Find the angles m and n at the top.



7. A highway crosses two parallel railway tracks. The acute angle between the highway and track 1 is 43°. Write the sizes of angles 5, 6, 7 and 8 at track 2 on the diagram. Give a reason for $\angle 6$ and for $\angle 5$.

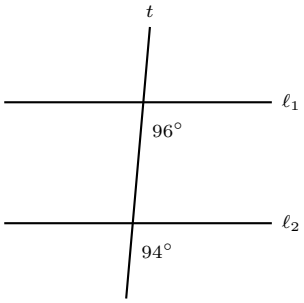


8. A farm gate has a top rail and a bottom rail that are parallel. The two end posts are perpendicular to the rails. A diagonal brace makes a 33° angle with the bottom rail.



(a) Find u, the angle between the brace and the top rail. (b) Find v, the angle between the brace and the post. Give your reason.

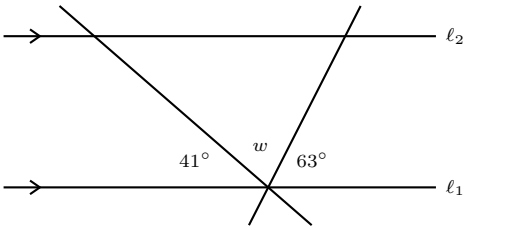
9. Sam put up two shelves, ℓ_1 and ℓ_2, along a slanted bracket t. He measures $\angle 3 = 96°$ and $\angle 7 = 94°$. He says, "Those are close, so my shelves are parallel."
(a) Is Sam right? Explain.



(b) Sam keeps ℓ_1 where it is. What should $\angle 7$ be? By how many degrees must he turn ℓ_2?

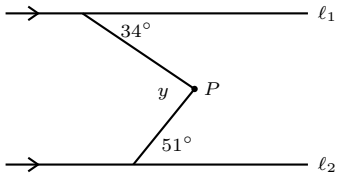
10. $\ell_1 \parallel \ell_2$. Which angle is always equal to $\angle 4$?
A. $\angle 5$ **B.** $\angle 6$ **C.** $\angle 7$ **D.** $\angle 3$

11. Numerical Response $\ell_1 \parallel \ell_2$. Two straight fences start at point A on ℓ_1 and make a triangle with ℓ_2. The angles at A outside the triangle are $41°$ and $63°$. The size of angle w, inside the triangle at A, is _____°.

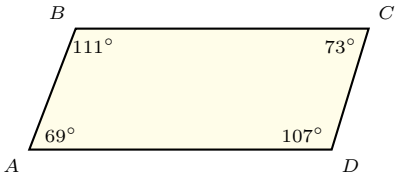


C. Stretch

12. $\ell_1 \parallel \ell_2$. A path bends at point P between the lines. Find the size of the bend, angle y.
Hint: draw a line through P parallel to ℓ_1. It splits y into two parts.

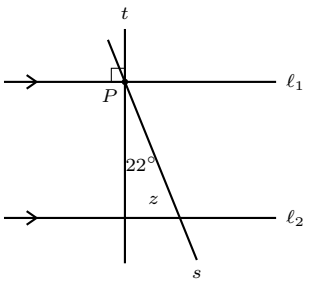


13. A sign is cut in the shape of quadrilateral $ABCD$. Its angles are $\angle A = 69°$, $\angle B = 111°$, $\angle C = 73°$ and $\angle D = 107°$.
(a) Which two sides are parallel? Explain.



(b) Is $ABCD$ a parallelogram? Explain.

14. $\ell_1 \parallel \ell_2$ and $t \perp \ell_1$. Line s goes through point P on ℓ_1. The angle between s and t is $22°$. Find z, the acute angle between s and ℓ_2. Give a reason for each step.



Extra Practice 9: Unit 5 Review

Practice Test 2. Treat this like the unit test. Show your work, and give a reason for every angle you find. A protractor, a compass and a ruler are allowed. In numbered diagrams, angles 1–4 are at the top crossing and 5–8 at the bottom crossing, clockwise from the upper left.

A. Skills

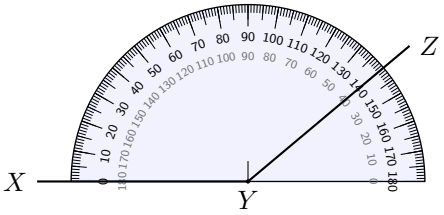
1. Fill in the blanks.

Angle	Type	Complement	Supplement
42°			
99°			

The reflex angle that goes with a 75° angle is _____.

2. Read $\angle XYZ$ on the protractor. Classify it, and explain why the answer is not 40°.

$\angle XYZ =$ _____ type: _____



3. Write each direction as a true bearing.

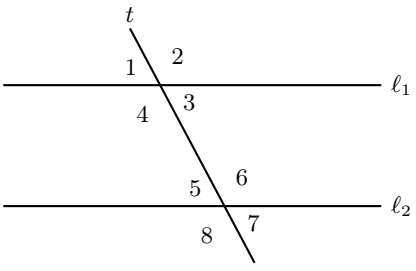
(a) due south _____

(b) 30° west of south _____

(c) 15° east of north _____

(d) the back bearing of 098° _____

4. Name each pair of angles.



(a) $\angle 4$ and $\angle 8$ _____

(b) $\angle 2$ and $\angle 8$ _____

(c) $\angle 3$ and $\angle 6$ _____

(d) $\angle 6$ and $\angle 8$ _____

(e) $\angle 1$ and $\angle 8$ _____

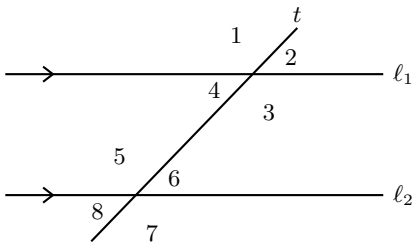
5. Polygons: the angle sum is $(n-2) \times 180°$.

(a) A six-sided garden has angles of 110°, 125°, 130°, 140° and 105°. Find the sixth angle.

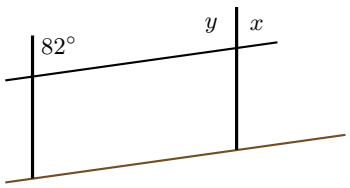
(b) Each angle of a regular polygon is 140°. How many sides does it have? (Try a few values of n.)

B. Applying

6. In the diagram, $\ell_1 \parallel \ell_2$ and $\angle 5 = 134°$. Find the other seven angles. Give a reason for each.



7. A fence is built on a hill. The posts are straight up, so they are parallel. The top rail slopes up the hill. At the first post, the rail makes an 82° angle with the post.

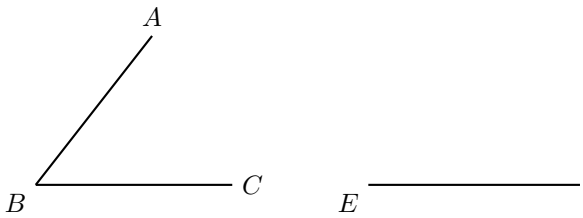


(a) Find x. Give a reason.

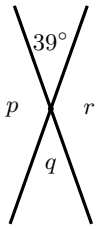
(b) Find y. Give a reason.

(c) Why can the builder cut the end of every rail at the same angle?

8. Copy $\angle ABC$ onto the ray from E using a compass and a straightedge. List your steps.



9. The two legs of a sawhorse cross like an X. The angle between the legs at the top is 39°. Find p, q and r. Give reasons.



10. Two lines are cut by a transversal, but the lines are **not** parallel. Which statement must be true?

A. Corresponding angles are equal.

B. Alternate interior angles are equal.

C. Interior angles on the same side add to 180°.

D. Vertically opposite angles are equal.

11. Numerical Response The sum of the interior angles of a polygon with 14 sides is _____°.

12. Race to 22. A token starts off the board, before square 1 of a strip numbered 1 to 22. Two players take turns moving it forward 1, 2 or 3 squares. Whoever lands on square 22 wins.

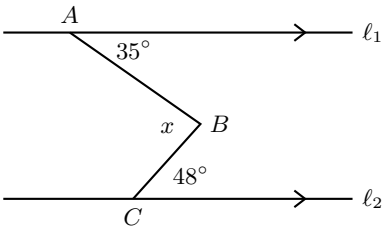
(a) Work backward. Which squares do you want to land on?

(b) Should you go first or second? What is your first move?

(c) Variation: the strip goes to 25, and each move is 1, 2, 3 or 4 squares. Should you go first or second now? Explain.

C. Stretch

13. $\ell_1 \parallel \ell_2$. Find x, the angle at the bend B. (Hint: draw a line through B parallel to ℓ_1.)



14. $\ell_1 \parallel \ell_2$. A transversal makes corresponding angles of $(2x + 10)°$ and $76°$.

(a) Find x. Check your answer.

(b) Find the interior angle on the same side of the transversal as the 76° angle.

15. A search-and-rescue team walks from base A on a bearing of 040° to a checkpoint B. At B they turn and walk on a bearing of 160° to point C. Sketch the route and find $\angle ABC$, the angle between the two parts of the walk at B.